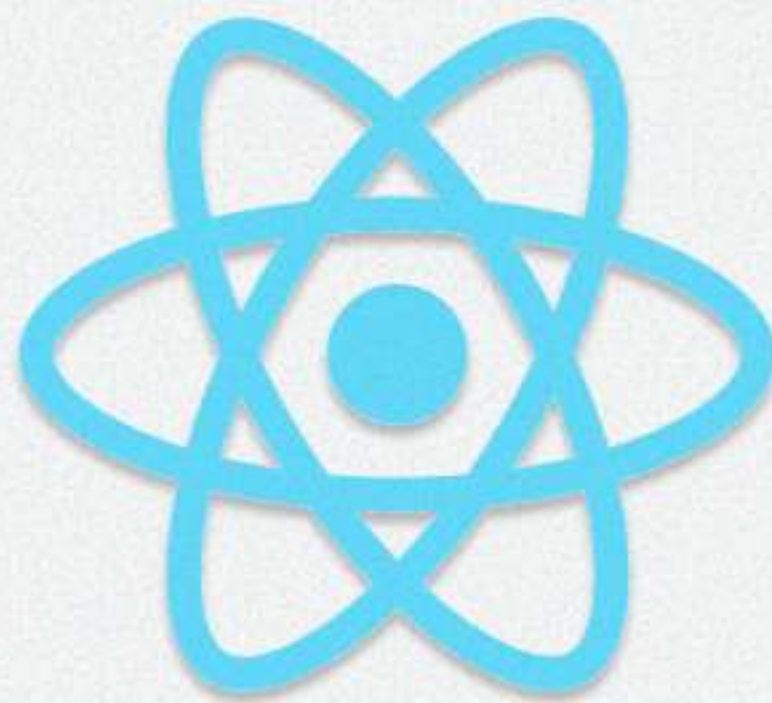


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React Rendering Concepts *Explained Simply.*

Understand how React decides what to render.



Initial Render

What is it?

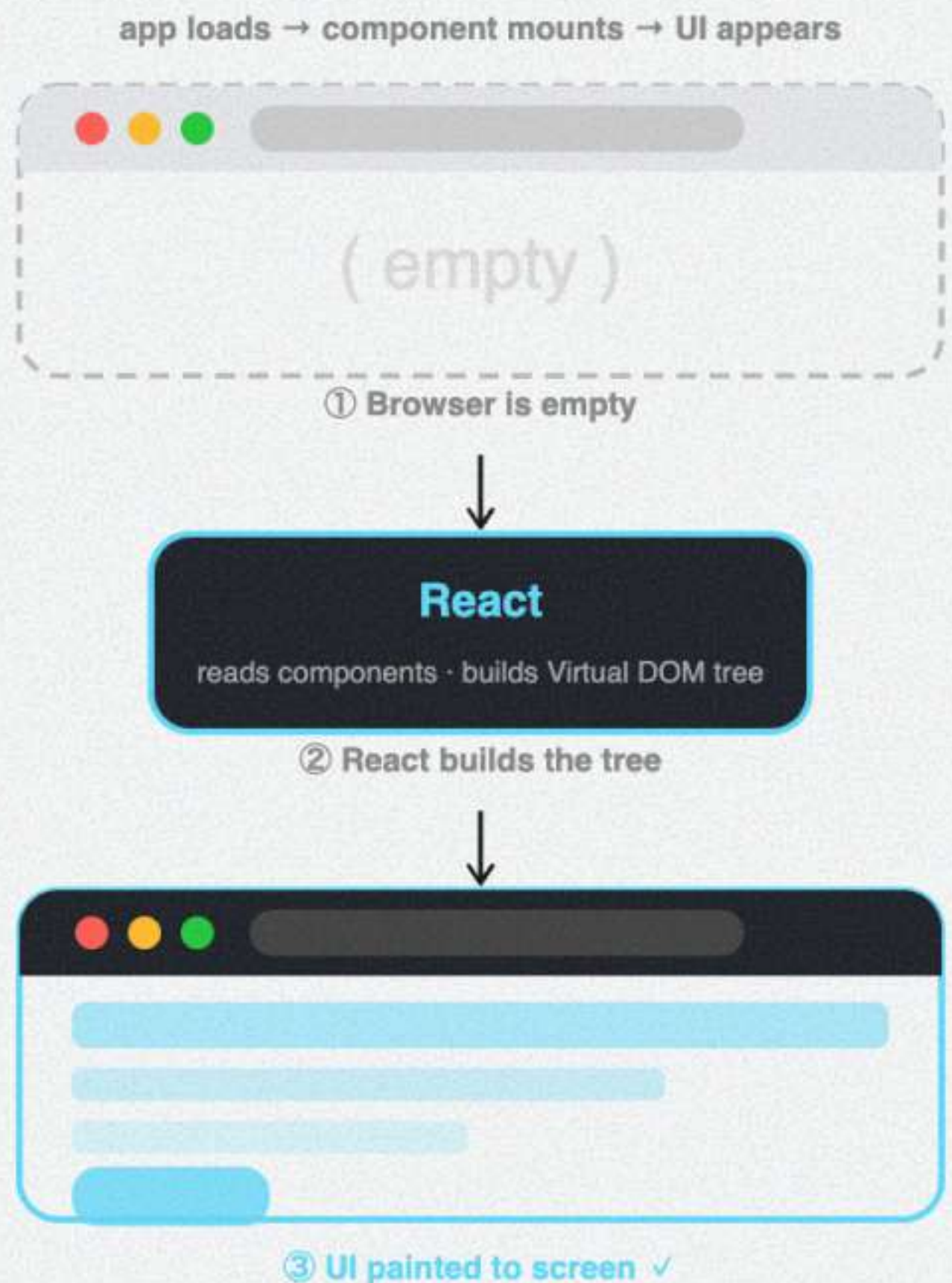
The first time React creates your UI and displays it in the browser.

Why does it matter?

Without the initial render, nothing appears on the screen.

Happens when

- App loads
- Component mounts



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Re-render

What is it?

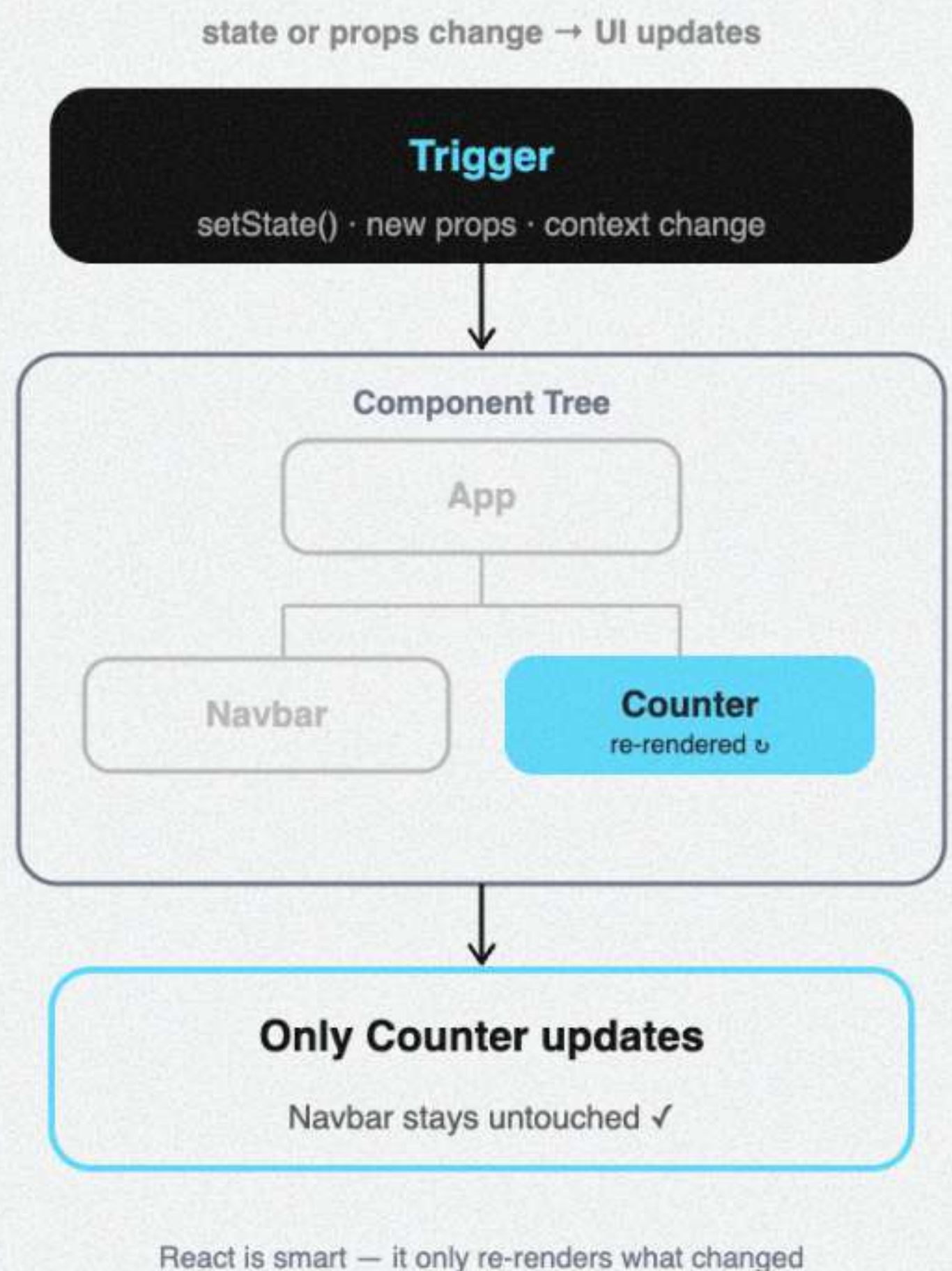
React renders a component again when its state, props, or context change.

Why does it matter?

Keeps the UI in sync with your latest data.

Happens when

- `setState()` is called
- New props are received
- Context value changes



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Virtual DOM

What is it?

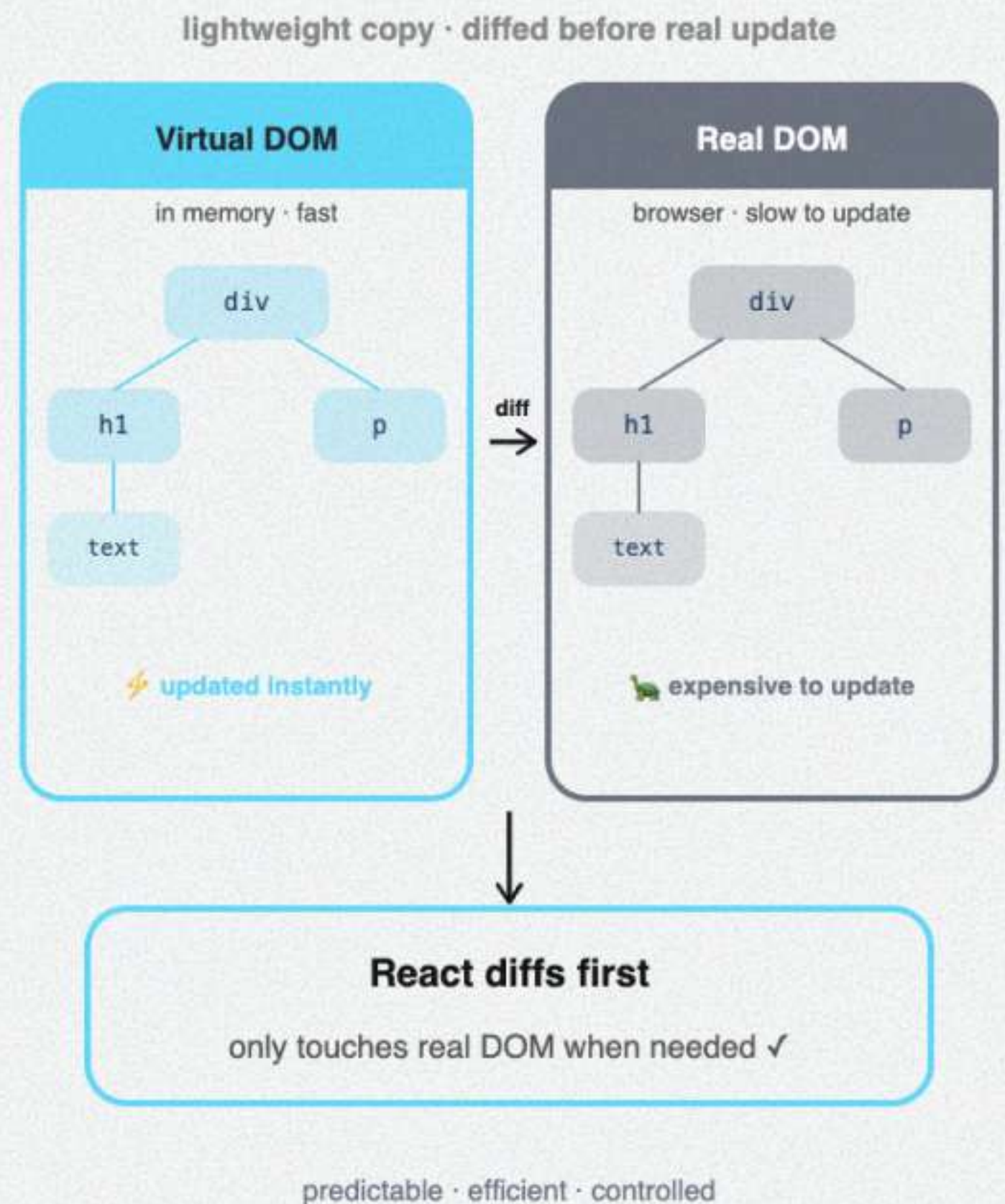
A lightweight copy of the real DOM that React uses to calculate UI updates.

Why does it matter?

React finds exactly what changed before updating the browser.

Benefit

- Faster and more efficient UI updates



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Reconciliation

What is it?

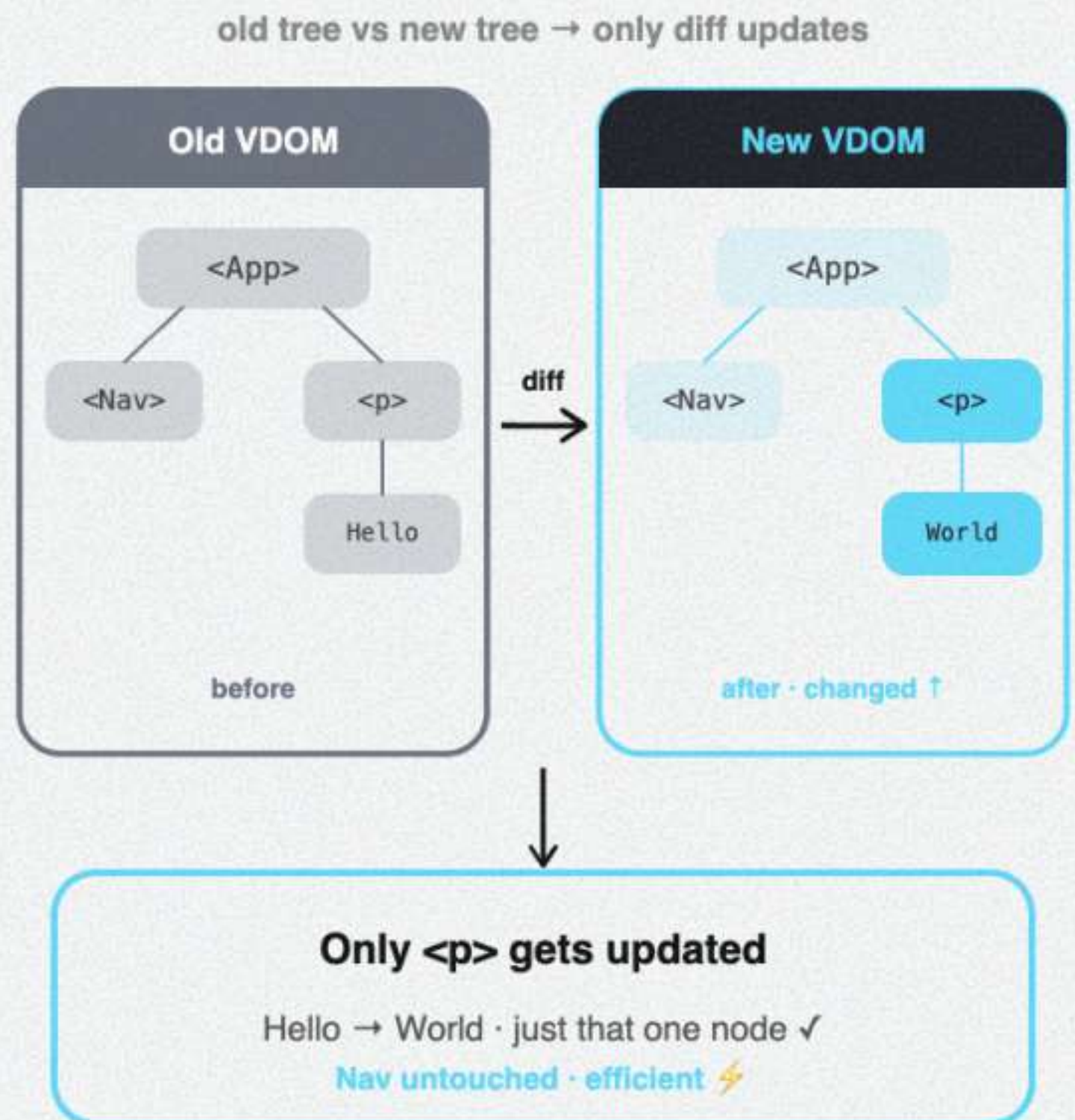
React compares the previous Virtual DOM with the new one to find the differences.

Why does it matter?

Only the changed parts of the UI are updated instead of rendering everything again.

Benefit

- Efficient rendering.



React's diffing algorithm at work

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Memoization

What is it?

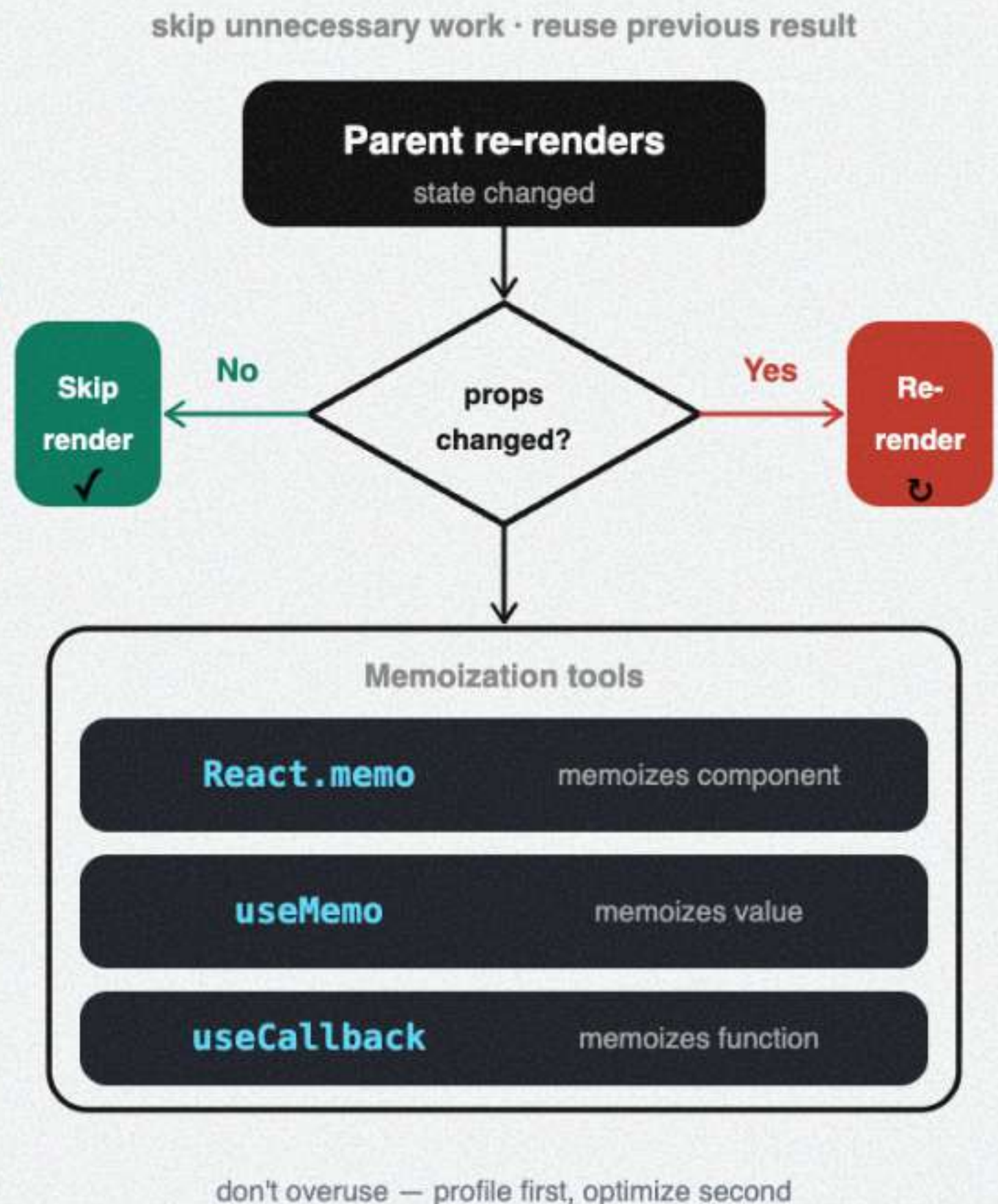
Stores previous results so React doesn't repeat unnecessary work.

Why does it matter?

Reduces unnecessary re-renders and improves performance.

Common tools

- `React.memo`
- `useMemo`
- `useCallback`



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Hydration

What is it?

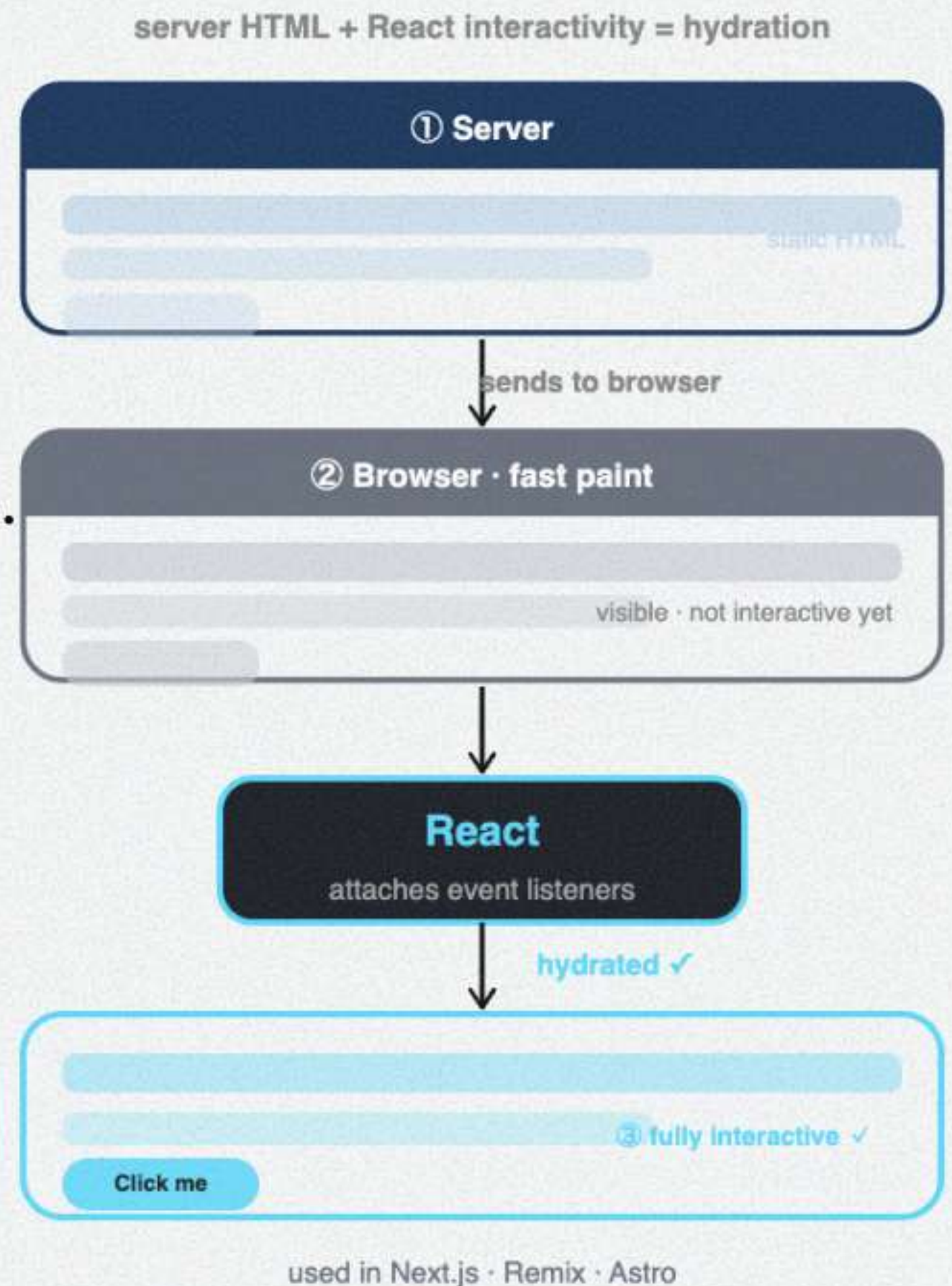
React attaches JavaScript and event handlers to HTML that was already rendered on the server.

Why does it matter?

Pages load quickly while becoming interactive afterward.

Commonly used in

- Next.js
- Remix
- Astro (React)



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Server Components

What is it?

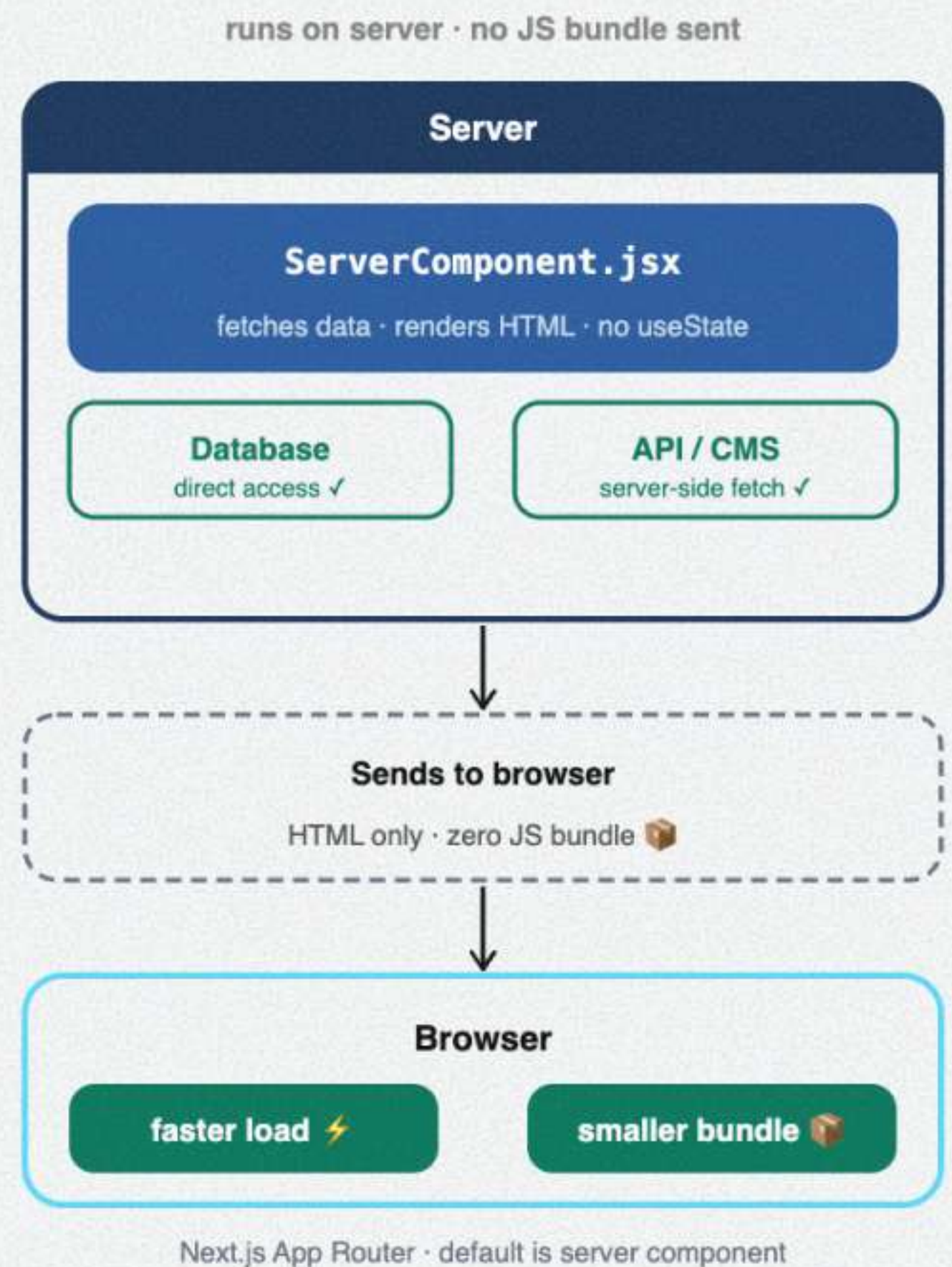
Components that run on the server instead of the browser.

Why does it matter?

They reduce JavaScript sent to the client and improve performance.

Best for

- Data fetching
- Static UI
- Faster page loads



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Client Components

What is it?

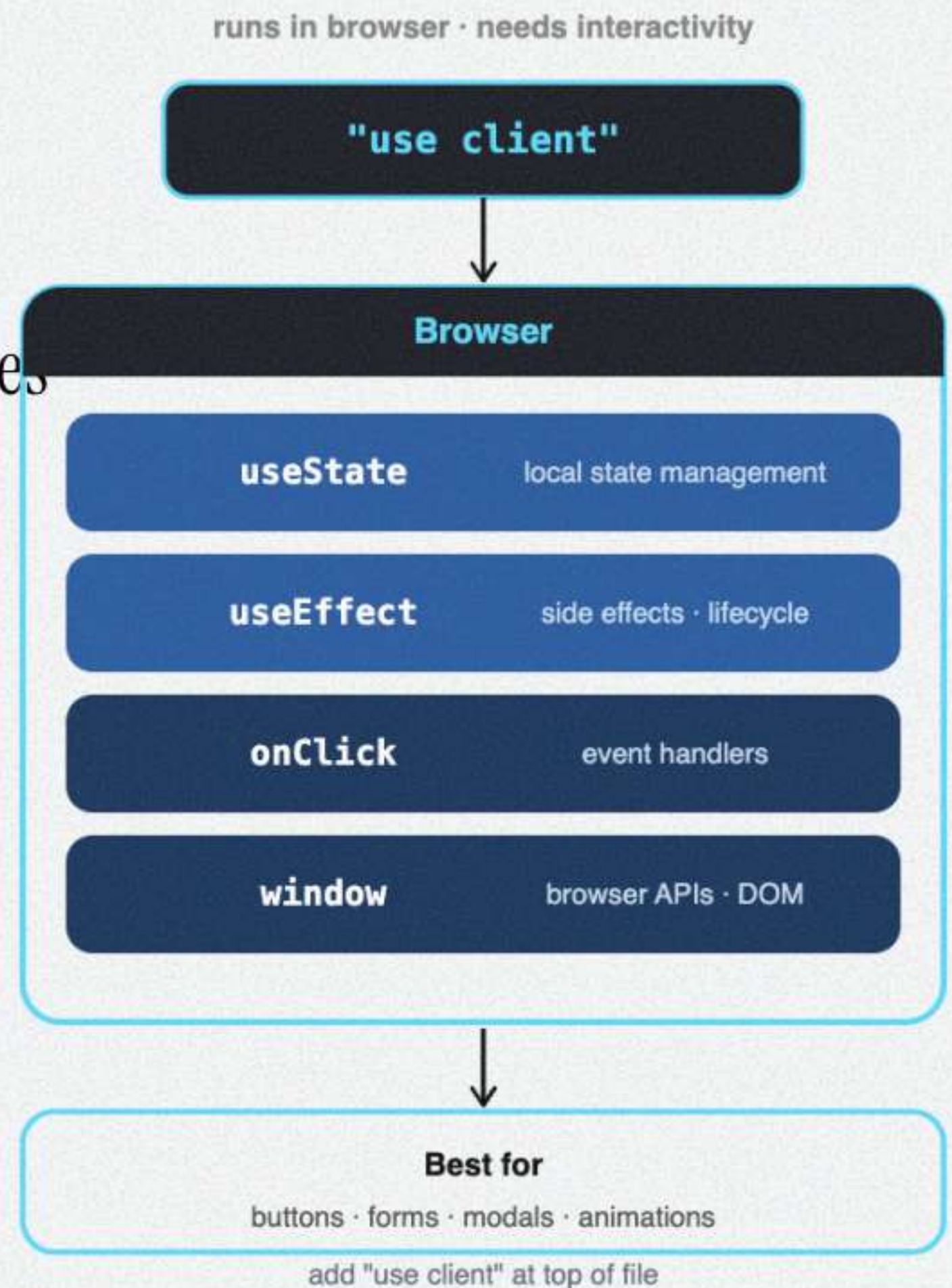
Components that run in the user's browser.

Why does it matter?

Required for interactive features like clicks, forms, animations, and local state.

Best for

- Buttons
- Forms
- Modals
- Interactive UI



React doesn't re-render
everything.

It carefully decides what,
when, and where to update.



Save this for your next
React project.



Comment “React” if you
found this helpful.